CHHAVI NAYYAR

(778) 636-0041 | chhavinayyar@gmail.com | linkedin.com/in/cnayyar | github.com/chhaviiiii | cnayyar.com

Education

University of British Columbia Sept 2022 – May 2027
BSc, Combined Major in Computer Science, Biology, & Cognitive Systems Vancouver, BC

- **Leadership:** Design Director – UBC UX Hub | TA – CPSC 532C (Human-Centered AI), COGS 300, COGS 303 | Member – Women in CS, Girls in STEAM

Industry Experience

Software Engineer Intern May 2026 – Aug 2026
Mastercard Vancouver, BC

- Built backend microservices in **Java** and **Python** with **gRPC/REST** for low-latency, real-time transaction processing on payments infrastructure
- Deployed and evaluated **ML pipelines** using **Docker**, **Terraform**, and **AWS** in production, supporting fraud detection with added test coverage and observability

Full-Stack Developer July 2025 – Sept 2025
UBC Michael Smith Laboratories Vancouver, BC

- Built responsive **TypeScript/React** site for the graduate Biochemistry program; established **CI/CD** and automated testing, cutting update turnaround by **30%** for 200+ stakeholders
- Conducted UX research; implemented WCAG 2.1 accessibility improvements across navigation and information architecture

Research Experience

Teaching Assistant – Human-Centered AI & Cognitive Systems Jan 2026 – Apr 2026
UBC Computer Science & Cognitive Systems Departments Vancouver, BC

- Supported graduate instruction for CPSC 532C/554C (Human-Centered AI), facilitating seminars on algorithmic fairness, bias, and evaluation methods
- Led lab sessions for COGS 300 and COGS 303, bridging cognitive science theory with applied AI implementation

Automation Engineer Sept 2025 – Apr 2026
UBC Extended Learning Vancouver, BC

- Built end-to-end automation pipeline in **Python** converting CSV survey data to structured PDF reports, eliminating **~70%** of manual processing
- Engineered automated notification workflows, reducing manual coordination overhead across course operations teams

Machine Learning Engineer June 2025 – Dec 2025
BC Cancer Vancouver, BC

- Engineered and evaluated unsupervised **ML** pipelines in **Python/scikit-learn** to analyze qualitative healthcare feedback, improving thematic pattern detection by **40%**
- Built and shipped an open-source concept-mapping toolkit (**NetworkX**, **FastAPI**) as a production backend service with interactive **Plotly** dashboards, cutting manual analysis time by **50%**; co-first-authored publication

Deep Learning Engineer Apr 2025 – Dec 2025
UBC Department of Computer Science Vancouver, BC

- Architected and evaluated VTNet models (**TensorFlow**, **PyTorch**) for real-time eye-tracking analysis, achieving a **25%** improvement in predictive accuracy
- Orchestrated distributed training experiments on **Slurm** HPC clusters, accelerating training pipelines by **30%** for large-scale iteration

Publications

- **Nayyar, C.***, Xu, H.*, Hilbers, D., Avery, J., Raman, S., Bates, A.T., Conati, C., Fayaz-Bakhsh, A., & Nunez, J. (2026). *Towards clinical implementation of artificial intelligence in cancer care: Concept mapping analysis of provincial workshop findings. Implement Sci Commun.* doi:10.1186/s43058-026-01075-x *co-first author

Technical Projects

Your Search Box – Multi-Modal AI Search Service | ***TypeScript, Next.js, Node.js, NLP*** 2025 – Present

- Built a server-side rendered, real-time search platform (**TypeScript**, **Next.js**) supporting voice, image, and text input, with an adapter pattern for headless CMS integration (Shopify, WordPress); live commercial product in production

Healthcare Concept Mapping Toolkit | ***Python, NetworkX, FastAPI, Plotly*** 2025

- Open-source backend toolkit for qualitative healthcare research: MDS, clustering, and thematic mapping from stakeholder input data
- Exposed via **FastAPI** endpoints with **Plotly** visualizations and automated evaluation reports; adopted by BC Cancer research teams

VR Recommendation System | ***Python, TensorFlow, Flask*** 2025

- Hybrid collaborative filtering and content-based engine (**TensorFlow**, cosine similarity) achieving 20–25% F1/precision gains; served via **Flask** REST API with batch inference and caching for high-frequency, low-latency requests

Technical Skills

- **Languages:** Python, Java, TypeScript, JavaScript, C++, SQL, HTML/CSS
- **AI/ML:** TensorFlow, PyTorch, scikit-learn, NetworkX, Plotly, ML pipeline evaluation
- **Backend & APIs:** Node.js, Express.js, Flask, FastAPI, gRPC, REST, Microservices
- **Cloud & DevOps:** AWS (EC2, S3), Docker, Terraform, CI/CD, Linux, Slurm
- **Frontend & Data:** React, Next.js, Angular, Tailwind CSS, PostgreSQL, MySQL, MongoDB, NoSQL